

Report of PhD activities in the academic year
2017 / 2018

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Abstract

This document describes the achievements, current state and future plans of the research project, moreover it contains the courses and conferences attended in the 2017 / 2018 academic year.

Contents

1	Current state of the research topic - Atmosphere modelling	4
1.1	Previous considerations	4
1.2	The model	4
1.3	Main result	4
1.4	Outlook	4
2	Courses, summer schools and seminars	4
2.1	Courses	4
2.1.1	Physics of the atmosphere - Prof G. Curci, Prof. G. Redaelli, University of L'Aquila	4
2.1.2	Numerical methods for hyperbolic problems - Prof. G. Russo (University of Catania), GSSI	5
2.2	Conferences, seminars, summer schools	5
2.2.1	Intensive Program on Fluids and Waves, GSSI	5
2.2.2	Women in applied and computational Mathematics, GSSI	5
2.2.3	Evans function methods in the stability analysis of periodic wavetrains, Prof. R. Plaza	6
2.2.4	On the spectral stability of traveling fronts for reaction diffusion-degenerate equations, Prof. R. Plaza	6
2.2.5	Secondment at the University of Sussex, Prof. Omar Lakkis	6
2.2.6	Waves in Flows Summer School in Prague	6

1 Current state of the research topic - Atmosphere modelling

1.1 Previous considerations

We had previously examined several different approaches to mathematically model both the atmosphere as a geophysical fluid, and a pollutant source in it as well. Some of the possibilities we considered were

- to use the classical 3D Cartesian coordinates (x, y, z)
- to use the monotonous relationship between altitude and pressure and use the (x, y, p) coordinates
- introduce the terrain influenced (x, y, η) coordinate system that includes height-related geographical diversity
- switch to spherical coordinate system and obtain the possibility to model globally

The model we decided to consider that is sufficiently simple to be mathematically manageable but at the same time meaningful first step.

1.2 The model

1.3 Main result

1.4 Outlook

2 Courses, summer schools and seminars

2.1 Courses

2.1.1 Physics of the atmosphere - Prof G. Curci, Prof. G. Redaelli, University of L'Aquila

This course gave an overview of selected topics related to the dynamics of the atmosphere. The general characteristics of the atmosphere were described in terms of mass and chemical composition. Gases with the highest mixing ratio are nitrogen, oxygen and argon - it is pointed out that these are constants, and the variabilities are provided by the remainder gases which are present in much smaller percentage: trace gases and aerosols. Although their lifetime is small, these later's effect is huge as they are able to capture radiation and alter energy balance. A big chapter in this course is represented by the thermodynamics of the atmosphere and gas laws. The ideal gas equation is considered in different forms, as well as the concept of hydrostatic balance and geopotential. The next step is the first law of thermodynamics in different approaches, and the notion of an adiabatic process. The main ideas and building blocks of chemical and photochemical reactions (thermal decomposition, photolysis) in the atmosphere

are introduced as well as the notion of the rate of reaction, and several chemical reaction equations are solved in this chapter.

2.1.2 Numerical methods for hyperbolic problems - Prof. G. Russo (University of Catania), GSSI

This course gave an overview on shock capturing schemes for the numerical solution of systems of conservation and balance laws. The emphasis was on the construction of high order finite volume and finite difference schemes, which are based on the three main ingredients: numerical flux, high order non oscillatory reconstruction, and high order integration in time. The first part of the course illustrated some models in which the nonlinearity produces shocks that propagate in time. Simple three point schemes, such as upwind, Lax-Friedrichs, and Lax-Wendroff was reviewed, and their behaviour on linear equations and systems analysed, before the illustration of more complex methods for the quasi-linear systems. The step by step construction of the schemes was illustrated in lab sessions, and the methods were applied to concrete examples of physical interest. The last part of the course was devoted to systems with source term.

2.2 Conferences, seminars, summer schools

2.2.1 Intensive Program on Fluids and Waves, GSSI

This training activity within the European Innovating Training Network Mod-CompShock surveyed recent results in the analysis of fluids and waves with emphasis on modelling and computational methods as well as its theoretical aspects.

- i) Week 1. Workshop/School in Numerics for Fluids and Waves
- ii) Week 2. Mini courses by J. Bedrossian. and D. Cordoba
- iii) Week 3. Mini courses by C. Bardos, Z. Grujic, J. Malek
- iv) Week 4. Workshop on Mathematical fine structures in fluid dynamics

2.2.2 Women in applied and computational Mathematics, GSSI

This three-day event was designed for researchers challenged by the application of various techniques in modern analysis and computational mathematics to intriguing problems arising in physics, material science and biology. The topic and the results of this present research project were represented - together with the work of many young female researchers - in the poster session.

2.2.3 Evans function methods in the stability analysis of periodic wavetrains, Prof. R. Plaza

This lecture series provided an introduction to recent techniques in the spectral stability analysis of spatially periodic travelling waves which involve the periodic Evans function introduced by Gardner.

2.2.4 On the spectral stability of traveling fronts for reaction diffusion-degenerate equations, Prof. R. Plaza

2.2.5 Secondment at the University of Sussex, Prof. Omar Lakkis

The aim of this 6-week secondment was to create both the theoretical and technical foundations for numerical simulations in order to visualise the results of the project. The main results and steps are the following:

- i) Software packages such as Docker, Anaconda, ParaView, FEniCS and Alberta have been installed
- ii) Writing scripts to login / connect to the HPC server at the Sussex University
- iii) Getting to know FEniCS: tutorial files and the FEniCS version of the "hello world" programs
- iv) Creating foundation for future work with specific theoretical details that arise when implementing this project - such as the loss of incompressibility when it comes to the FEM approach, and the use of Lagrange multipliers.

2.2.6 Waves in Flows Summer School in Prague

The aim of this summer school was to present a comprehensive series of lectures on various aspects of waves and their role in mathematical analysis and numerical simulation of different types of flows. The school was organised as a multidisciplinary event with topics ranging from theoretical and numerical mathematics up to physics and applications. The emphasise was on models related to topics originating in environmental, biomedical and industrial applications. Besides of the course lectures, the program included a one day workshop dedicated to short lectures given by attending scientists.

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